BEYOND EVALUATION USE

An Empirical Study of
Involvement and Use/Influence in
Multi-Site Evaluations

Report 5:
A Case Study of the Impact and Involvement of the Utah State Math Science Partnership (MSP) – Research, Evaluation, and Technical Assistance (RETA) Project

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Executive Summary

In November 2004, the National Science Foundation (NSF) awarded Drs. Frances Lawrenz and Jean King of the University of Minnesota a grant ("Beyond Evaluation Use") to study the relationship between involvement and use in multi-site evaluations. The overall question guiding this research is: What is the relationship between involvement in evaluation and use of the evaluation processes and outcomes in multi-site evaluations? The grant activities were aimed at describing and measuring how four NSF-funded program-level evaluations involved projects in the multi-site evaluation and, in turn, how this level of involvement affected the ways in which individuals, the projects, and the science, technology, engineering, and mathematics (STEM) education and evaluation communities used the process or results. NSF was the primary "client" for the evaluation. Therefore, the individual projects and the fields of STEM education and evaluation were secondary audiences, not directly the intended users.

Evaluation projects related to the following four NSF programs were included in the study: Advanced Technological Education (ATE), Collaboratives for Excellence in Teacher Preparation (CETP), Local Systemic Change through Teacher Enhancement (LSC), and Mathematics and Science Partnerships – Research, Evaluation and Technical Assistance (MSP-RETA).

Data were collected through an online survey distributed to evaluation project leaders[1] and evaluators, interviews with project leaders and evaluators, an interview with one of the NSF program officers responsible for the program at the time, an on line survey of NSF education project PIs and questions of journal editors. Additional data were collected through review of archival documents, publications, and reports produced by the evaluation projects and through reflective descriptions provided by the evaluation projects principal investigators.

This case study examines the relationship between involvement and use in the program-level evaluation of one of these programs – the Utah State MSP-RETA evaluation project. Findings suggest that the MSP-RETA project had a moderate level of influence on the MSP projects, which was only slightly enhanced by the involvement of the projects in the evaluation technical assistance. This case study presented evidence of influence of the program evaluation on both the projects and the field. Other variables that affected use included project perceptions related to the extent to which the evaluation was representative of their own project and the opportunities the program provided for interaction and collaboration with other projects and evaluators.

The case study showed a weak relationship between project involvement in the MSP-RETA evaluation project and the influence of the evaluation's processes and products on the field. The weak relationship between involvement and use was primarily due to the time in the project life cycle at the time of this research.

Introduction

In November 2004, the National Science Foundation (NSF) awarded Drs. Frances Lawrenz and Jean King of the University of Minnesota a grant ("Beyond Evaluation Use") to study the relationship between involvement and use in multi-site evaluations. The overall question guiding this research is: What is the relationship between involvement in evaluation and use of the evaluation processes and outcomes in multi-site evaluations? The grant activities were aimed at describing and measuring how four NSF-funded program-level evaluations involved projects in the multi-site evaluation and, in turn, how this level of involvement affected the ways in which individuals, the projects, and the science, technology, engineering, and mathematics (STEM) education and evaluation communities used the process or results. NSF was the primary "client" for the evaluation. Therefore, the individual projects and the fields of STEM education and evaluation were secondary audiences, not directly the intended users.

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This case study examines the relationship between involvement and use in the program-level evaluation of one of these programs – the Utah State MSP-RETA evaluation project. The case study has three sections. First, it describes the history of the MSP-RETA evaluation project and the context within which the project was conducted. Second, it examines the factors related to the project's involvement, influence, and use, specifically, the extent to which the MSP projects were involved in the RETA and used the Utah State MSP-RETA activities and products. And finally, it describes the extent to which the MSP-RETA evaluation project influenced the fields of STEM education and evaluation.

Methods

Data were collected through interviews with MSP project leaders and evaluators and an interview with the NSF program officer responsible for the MSP program at the time. In addition, data were collected by review of archival documents, publications, and reports produced by the MSP Program and the Utah State MSP-RETA Project and through the Utah State MSP-RETA principal investigator's (PI's) reflections. The case study also utilized findings based on the responses to open-ended questions from an online survey distributed to MSP project leaders and evaluators (n=56), as well as results of a survey of actively-funded NSF program PIs (n=191).

MSP Project Leader and Evaluator Survey

The research team acquired names and email addresses for the PIs, co-PIs and evaluators of MSP projects. The team compiled lists from project and meeting rosters and online NSF records. When e-mail addresses were not provided for individuals on the list, a member of the team searched the internet for an email address. In all, the research team acquired names and email addresses for 165 MSP project leaders and evaluators.

Each individual received an initial invitation email with a description of the survey, an explanation of its use, the Human Subjects approval number, and telephone numbers to call with questions. The email also contained al ink to the online survey with a unique login and password. Two and four weeks following the initial email, reminder messages were sent to nonrespondents. The survey data were collected from August 10, 2006 through January 17, 2007.

Complete details of the survey's development, administration, and validation are presented in "Beyond Evaluation Use Report 1: Findings from the NSF Follow-up Survey." Survey findings presented in this report are taken from 57 close-ended items and four open-ended questions related to the involvement in and use of the ATE program evaluation.

Fifty-six people responded to the survey (34% response rate). Four people were unreachable. In terms of the respondent roles, 3 evaluators (5.4%) and 53 non-evaluators (PIs, project staff, and other) responded.

The survey had some limitations due to the wording of questions. The research team decided to keep questions consistent as much as possible across all four program evaluation projects, including MSP-RETA. Thus, a number of questions included "MSP-RETA program evaluation" wording, as well as references to program evaluation "planning," "implementation," and "findings." However, as explained in the introduction, the Utah State MSP-RETA did not conduct a distinct program-level evaluation. Rather, as a project, the Utah State MSP-RETA promoted and assisted with evaluation for all other MSP projects. Therefore, a number of survey items were not appropriate for MSP-RETA.

Responses to the survey's open-ended questions and the follow-up interviews with the survey respondents suggested that a number of study participants had correctly understood the meaning of the survey questions. Nevertheless, a number of survey respondents in their responses to the open-ended questions and in the individual email messages indicated that they did not understand the questions and were confused about what they were referring to. For example, one respondent noted that his reading of the survey questions suggested that there had been a program evaluation of the MSP-RETAs, whereas he took no part in such an evaluation and was unaware of anything about it. He expected to be asked questions about the technical assistance

provided through the MSP-RETA on building evaluation capacity. Another person believed that the survey questions did not apply to her despite the fact that she participated in a few MSP-RETA activities.

The above-mentioned limitations suggested that there was the possibility of a bias of unknown magnitude in the data collected through the survey. Therefore, the research team has exercised caution in analyzing and interpreting the results of the survey, as well as in making conclusions. It was decided to exclude the quantitative results of the survey from the analysis while keeping unambiguous responses to some of the survey's open-ended questions as part of the qualitative analysis.

Interviews

From March through April, 2007, we conducted individual telephone interviews with four project leaders from different MSPs who completed our survey (the Project Leader and Evaluator survey) regarding their experiences with the Utah State MSP-RETA and indicated their willingness to participate in a follow-up interview. As suggested by the Advisory Committee, we conducted additional individual telephone interviews with four more MSP project leaders in March, 2008. We used purposeful and maximum variation sampling, thus we purposefully selected participants from the pool of volunteers to include leaders from various MSP projects.

In the first stage, we attempted to purposefully select a sample of four project leaders and evaluators based on their responses to the involvement and use questions in the survey that were used to identify four key types of respondents: 1) high involvement, high use; 2) high involvement, low use; 3) low involvement, high use; and 4) low involvement, low use. These categories were determined based on relative reported levels of involvement and use. In the second stage, four more individuals were sampled who, according to the survey results, were both highly involved in the evaluation and also reported high use of the evaluation. However, due to the survey's known limitations, it was eventually decided that the use of the above categories was not realistic for the analysis of the MSP-RETA interviews.

Potential interviewees were contacted by email and asked to participate. We were interested in hearing their perspectives on how the Utah State MSP-RETA project affected them and the ways in which they may or may not have been involved in its activities. The interviews followed a standardized, semi-structured protocol with flexibility in probing for additional details and description as necessary. On the average, the length of the interviews was approximately 35 minutes. All interviews were taped and transcribed.

One additional interview was conducted with an NSF program officer responsible for the MSP program. This interview lasted approximately 25 minutes and was also taped and transcribed.

Citation Analysis

The citation analysis used bibliographic methods to gather data about the use of program evaluation products by individuals within the STEM education and program evaluation communities. Citation analysis consists of tracking the number of citations other authors make to a published work. Web of Science, Google Scholar, and Google were used to collect citation information from lists of evaluation products produced by the MSP-RETA program evaluation project (see Appendix A for a list of the MSP-RETA evaluation products).

The list of the MSP-RETA evaluation products was initially compiled by a University of Minnesota team member. The list was then forwarded to Horizon Research Inc. (HRI) staff members who reviewed the list and made changes. The MSP-RETA program evaluation project PI was then provided with the opportunity to review the final list. Products were defined as any publications, presentations, instruments, or other related materials that were produced as part of the NSF-funded evaluation project.

The Web of Science, Google Scholar, and Google search engines were used to collect citation information from the evaluation product list. Searching was completed in late 2007 during the following periods: Web of Science October 23-25, Google Scholar October 25-November 13, and Google November 13 – December 2.

Data were collected manually by entering full reference information about the citations into an Excel spreadsheet which was later checked for accuracy and compliance with APA style and converted into an SPSS data file. Complete details of the rationale, development, administration, and validation of the citation analysis are presented in "Beyond Evaluation Use: A Citation Analysis of the Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education and Evaluation Fields, Report 6."

NSF PI Survey

As a measure of influence on the field, the research team administered an online survey to a sample of 350 individuals randomly selected from the database of over 2,000 currently-funded NSF Education and Human Development Directorate Principal Investigators (PIs). The survey elicited responses about PIs' perceptions of evaluations, people, and publications that might have been influential on the fields of STEM evaluation and education. The overall response rate was 54.7%. Complete details of the survey's development, administration, and analysis are presented in "Beyond Evaluation Use Report 7: Findings from the Survey of Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education Field."

Journal Editor Inquiry

A very brief email/telephone survey of editors and coeditors of six journals in the evaluation and STEM education fields was also conducted to determine if the editors were aware of any articles that had been published about influential STEM evaluations. Responses were obtained from three of six editors/coeditors. Complete details of the survey's development, administration, and analysis are presented in "Beyond Evaluation Use Report 7: Findings from the Survey of Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education Field."

Archival Review

The collection of archival documents for MSP was primarily conducted using electronic databases and Internet search engines. The NSF website provided information on the MSP program as well as on the MSP-RETA Project at Utah State University. The NSF documents included reports, grant solicitations, project summaries, and a "dear colleague" letter.

The primary source of information was the NSF-supported "MSPnet," which is an electronic learning community that offers a centralized location for information on MSP projects, links to several other program-related websites, and the "RETA Showcase" that provides links to a variety of presentations given by project personnel. We also searched using Google, several electronic indexes including Academic Search, WilsonWeb (the H.W. Wilson Information Retrieval System for the World Wide Web), ERIC (the Education Resources Information Center), and the NSF-sponsored OERL (the Online Evaluation Resource Library). Complete details of the document collection and analysis are presented in "Beyond Evaluation Use Report 6: A Citation Analysis of the Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education and Evaluation Fields."

Utah State MSP-RETA Background and Context

This section of the case study presents details of both the MSP Program and the Utah State MSP-RETA evaluation project to provide context for discussion of evaluation involvement and use.

MSP Program

The NSF Math Science Partnership Program (MSP) began as a presidential initiative in 2001 and was authorized by Congress in 2002. [2] MSP is a major research and development effort that supports innovative partnerships to improve K-12 math and science instruction, enhance student achievement, and reduce the achievement gap in math and science performance among diverse student populations.[3]

MSP supports the development of partnerships to integrate the work of higher education and K-12 STEM faculty. Typical project activities include new curriculum development and skill-building for teachers. Successful projects serve as models that can be widely replicated. [4] The partnerships must support the programs' goals to provide challenging coursework; increase the quantity, quality, and diversity of math and science teachers; and establish results-oriented projects that implement evidence-based educational practices.[5] In 2006, MSP was bringing together 150 institutions of higher education, 550 school districts, and 3,300 schools, along with corporate partners including Pfizer, Ford, Texas Instruments, Inc., Xerox, IBM, Merck, and Intel.[6]

MSP awards competitive grants to teams of institutions: higher education, K-12, and other partners.[7] MSP projects emphasize innovation to improve K-12 math and science education. Since 2002, the MSP Program has made 80 awards totaling over $500 million across four program types: [8]

1. Comprehensive projects implement change across the K-16 continuum;
2. Targeted partnerships affect a narrower grade range or disciplinary focus;
3. Institute Partnerships build partnerships between university faculty and K-12 teachers[9]; and
4. Research, Evaluation and Technical Assistance (RETA) projects develop tools to build research, evaluation, and infrastructure capacity for the MSP.

Evaluation Capacity Building (ECB) in STEM

"The field of evaluation is…facing a shortage of evaluators with a STEM background who can evaluate programs that require STEM expertise. The National Science Foundation (NSF) has placed priority on developing STEM evaluators who have both a strong STEM background and an understanding of educational contexts.
…Previous ECB efforts…have created training opportunities in STEM evaluation, but the field of STEM evaluation needs a more comprehensive model of ECB. Overall, there is a need for evaluation training programs to move beyond a training view of evaluation capacity building, to include long-term activities and experiences that can immerse… stakeholders… in evaluation. It is also important to foster collaborative experiences and not just individualized experience, especially since the practice of evaluation is often highly collaborative. This involves workshops, education, and institutes, along with immersion in real-world internships."

From: Huffman, D., Lawrenz, F., Thomas, K., Clarkson, L. (2006). Collaborative Evaluation Communities in Urban Schools: A Model of Evaluation Capacity Building for STEM

Utah State MSP-RETA Overview

The RETA projects use MSPs as research sites to improve K-12 STEM education. The RETA projects conduct rigorous, innovative projects and provide technical assistance to MSP projects, particularly in the area of data use. Of the more than 30 MSP-RETA projects, this case study considers just one: the Utah State University MSP-RETA project, which is titled "Building Evaluation Capacity of STEM Projects-MSP" and which was established in October 2002. The Project Director and PI of the Utah State RETA is Catherine Callow-Heusser. Unlike almost all the other

MSP-RETA projects, the Utah State RETA received a grant of $1.5 million over three years, while most received only one year of funding.

The purpose of the Utah State RETA is not to evaluate MSP projects, per se. Notwithstanding the original intention that it would gather data to inform an overall program-level evaluation, this RETA was designed to work with the MSP projects to identify evaluation needs and develop program evaluation models based on those needs. [10] Further, this RETA offers technical assistance and expert consultation via a network of evaluation consultants and supports the development of professional communities of teachers, researchers, evaluators, cognitive scientists, and social scientists and others who depend on evaluation for continued improvement and for generating new knowledge.[11]

Utah State MSP-RETA Activities

The Utah State RETA provided technical assistance through regional workshops and meetings and by using a network of nationally-recognized evaluation consultants who provided direct assistance in evaluation planning and implementation to MSP projects. The Utah State RETA focused on evaluation planning and capacity building to maximize the value of MSP evaluation activities.

The Utah State RETA Building Evaluation Capacity project provided evaluation technical assistance to individual MSP projects. Evaluation questions and designs were determined in collaboration with project staff to meet local contextual needs. The Utah State RETA project also worked on developing state-of-the-art evaluation models for STEM education projects.

ECB Efforts in STEM by NSF

"…1. Advance research on science, technology, engineering, and mathematics education and improve evaluative research on STEM education programs;
2. Increase the capacity of the field to conduct high-quality, innovative, useful, and credible STEM education evaluation and research studies; and
3. Increase the capacity of STEM education researchers and evaluators to communicate the results of their research.

…Support projects designed to enhance the capability and infrastructure of the education field to conduct evaluations through education and training, the development of evaluation knowledge and skills, and through the creation of evaluation resources useful for the field in general."

From: National Science Foundation. (2005). *Evaluative Research and Evaluation Capacity Building (EREC) and Research on Learning and Education (ROLE).* A solicitation of the Division of Research, Evaluation, and Communication (REC) Program Solicitation, NSF 05-529. Available online at http://www.nsf.gov/pubs/2005/nsf05529/nsf05529.htm.

Based on a request of the NSF and to serve the need for evaluation design assistance, the Utah State RETA developed and disseminated the Design-Implementation-Outcomes (DIO) Cycle that incorporated elements of evidence-based evaluation design. The DIO Cycle provided a framework for considering project activities at multiple levels and from

multiple perspectives. The Utah State RETA staff intended that evaluations designed with input from the DIO Cycle would provide evidence for accountability, improve projects, and support the value and feasibility of projects and activities.[12]

In addition to the DIO Cycle, the Utah State RETA developed a short course on evaluation targeting specific MSP stakeholders – public school teachers – as well as school administrators, parents, and non-evaluators. The goal of the short course was to give these individuals an idea about the activities of the evaluators.[13] The Utah State RETA also developed an online logic model tutorial and a database of online evaluation resources for use by local project staff.

NSF Program Officer Perspectives on the MSP-RETA

This section provides one NSF program officer's thoughts about NSF's involvement in the Utah State MSP-RETA and how individual MSPs were involved. It also describes uses of the Utah State MSP-RETA and its influence on NSF, the MSPs, and the community, as well as the relationship between involvement and use as perceived by the NSF officer. It must be emphasized that this section reflects perspectives of only one NSF officer.

NSF Involvement. The program officer was fairly heavily involved with the Utah State MSP-RETA. For example, she interacted with the PI, made suggestions for what the project should do, attended meetings the project sponsored, and encouraged submission of supplements. She felt, however, that she did not "control" the project or influence it unduly. As the emphasis on rigorous evaluation became more explicit in the MSP program, more MSP partnerships turned to the Utah State MSP-RETA for support. Because it was a grant and not a contract, they did not have to report to NSF on anything that might have been wrong with the projects. Reportedly, NSF also deliberately stayed out of the process in terms of commenting on the Design-Implementation-Outcomes (DIO) Cycle booklet during its development, because NSF wanted it to be the community's product.

Many MSP projects were involved in the Utah State MSP-RETA. However, NSF was cautious at the beginning about recommending involvement because they thought the Utah State MSP-RETA might overwhelm MSP staff, who were busy with their program activities.

Influence on NSF. The interviewee's perception was that the Utah State MSP-RETA had positive influence on all three audiences, perhaps most strongly on the community that the MSPs formed. Because the Utah State MSP-RETA activities were so comprehensive, an influence on the MSP community was more akin to influence on the STEM community. The Utah State MSP-RETA activities were inclusive of all the MSPs toward the end, so that means they possibly in some ways affected all of the MSPS, and then, with the MSPs being comprehensive, the Utah State MSP-RETA affected the STEM community.

Reportedly, the NSF staff "learned at those meetings a lot about evaluation, a lot about our projects." There was also a realization of a need for a good evaluation:

> …the question was whether or not we had strong evaluation. They [projects] accepted it [DIO], and their concern had been before that we were not paying serious attention to evaluation.

NSF Perceptions about Influence on the Projects. The Utah State MSP-RETA influenced MSP projects by helping them with learning about, planning, and implementing their evaluations. For example, Utah State MSP-RETA helped one MSP project "to change their evaluation and really work much harder and… have a much, much stronger evaluation plan than they did before." As a result of the Utah State MSP-RETA work, those who took advantage of it improved their evaluation activities. The project also helped develop a community that is "focusing on evaluation at a much higher level."

NSF Perceptions about Influence on the Field. The interviewee believed that the field would also learn a lot especially if the projects involved with the Utah State MSP-RETA started publishing some of their findings. For example, the NSF distributed one of the Utah State MSP-RETA documents, the DIO document, by giving it out at other places.

She also noted personal contributions of the Utah State MSP-RETA staff, as well as of experts they involved in assisting MSP projects:

> You know, it's hard to pick out any one thing. But by bringing together experts outside and experts inside and talking, I think the evaluators probably learned a lot because you always do when you are working with something as complex as an MSP.

From the program officer's perspective, both the MSP projects and the STEM community reportedly benefited from the Utah State MSP-RETA's work. The people in the MSP learned more about what evaluation is and what needs to be done. While the individuals in MSP projects learned, the evaluation capacity of the whole community improved.

NSF Beliefs Regarding the Relationship between Involvement and Use. The interviewee felt that a more active pattern of involvement would lead to more use of the information. As she observed,

> Those people who actually sat and talked with evaluators and tried to struggle through the really hard thing of how I'm going to possibly tease out the impact of my intervention – a very complex world and outcome – are going to have learned a great deal.

Utah State RETA PI's Perceptions of Involvement, Use, and Influence

The Utah State MSP-RETA's PI, Catherine Callow-Heusser, also provided her opinion on the level of the projects' involvement in both the planning and implementing of the Utah State MSP-RETA evaluation activities. According to her, the Utah State MSP-RETA staff was not involved in evaluating projects or conducting the evaluation work. The Utah State MSP-RETA team and the expert consultants they contracted for MSP projects mostly provided evaluation technical assistance (i.e., recommendations about evaluation design and methods, statistical methods, evaluation instruments, holding meetings for PIs and staff and evaluators to learn more about evaluation, etc.).

Paraphrasing her reflections[14] [15], the Utah State MSP-RETA's PI felt that the project had evolved over time. According to the initial proposal, the MSP-RETA was intended to provide evaluation design consultation. However, it appeared that most MSP projects already had evaluations designed, so the MSP-RETA refocused to provide evaluation technical assistance and selected 10 projects that had a design in place which the MSP-RETA staff thought they could strengthen. As Callow-Heusser stated, many of those projects "closed the doors on us" because they already had evaluations in place and did not want to change what they were planning to do. Eventually, it was projects in the middle of the spectrum – with competent evaluators and fairly good evaluation designs – who actually sought out and utilized the MSP-RETA's help.

The MSP-RETA provided the MSPs with actual hands-on data collection and analysis-related technical assistance. Over time, as the project staff became more familiar with the needs of the MSPs, the technical assistance provided became more related to helping projects implement evidence-based decision making, particularly as they made mid-course corrections. The project also moved toward providing opportunities for the MSPs to interact and share their problems and solutions related to evaluation. In other words, the Utah State MSP-RETA provided direct influence through the involvement of evaluation technical assistance consultants or indirect influence through larger workshops and meetings. This indirect influence resulted in the development of the Design-Implementation-Outcomes Cycle, a reportedly important tool for the MSPs. In summary, the project's influence occurred mainly through involvement during various phases of evaluation planning and implementation.

Project-Level Involvement and Use

This section of the case study examines the relationship between project involvement in the program evaluation project and the use of the evaluation processes and outcomes. It presents data from the eight MSP project leaders and evaluators' interviews and, partially, from responses to the survey's open-ended questions. The section will report the study of participants' perceptions of their involvement in and influence of the Utah State MSP-RETA. The following findings are organized according to the following topics:

- Motivation for Involvement

- Levels of Involvement
- Perceived Effect of Involvement
- Use by Project Leaders and Evaluators
- Use by Projects
- Factors Affecting Use

Motivation for involvement

There were a few things that motivated people to become involved with Utah State MSP-RETA activities. For example, two interviewees mentioned their *desire to improve* and be effective:

> …we wanted to make sure that we took every opportunity that was offered by NSF through this RETA project to learn as much as we could about that and to be as effective as we could within our project. (MSP Interviewee 1)

> My own personal interest of trying to improve on the evaluation of the project that we have kind of motivated me to get involved and learn what other projects were doing. (MSP Interviewee 3)

The *lack of evaluation capacity* in their projects was a motivating factor for another five participants. Those projects had a severe lack of evaluation expertise and experience and needed some external guidance and assistance on developing their capacity, as well as plans for evaluating their programs.

> …Our project has limited evaluation capacity…it was that internal part where we had little expertise in evaluation, so we were looking for any resources we could find. This one seemed like a good fit that would be good technical assistance for our project where we had limited existing capacity. (MSP Interviewee 2)

> We had very little background on educational evaluation. …We needed outside help. We also wanted to try our ideas out on some people who have perhaps a national view to see if maybe some ideas that we thought were good could be supplemented or modified or in some cases, replaced with better ideas. (MSP Interviewee 5)

One person reported that one of the requirements for his MSP was that the project be evaluated, and it prompted him to engage in the Utah State MSP-RETA activities. Another respondent mentioned her *interest in other projects' work*:

> I think for me since I'm Director of Research and Evaluation, I'm always interested in finding out what other projects are doing to try to get information, some of the problems they have, some of the successes they have. So the RETA's

> opportunity allowed me the opportunity to basically communicate… (MSP Interviewee 3)

According to one PI, some of his Co-PIs had been at meetings where the various RETAs were making presentations. They were *impressed by the Utah State RETA*, and that was why they called on them. (MSP Interviewee 4)

For another interviewee, MSP-RETA was "a vehicle for dissemination which was something we promised NSF." She emphasized that "a very important part of our goal in our project is to disseminate our findings, and these meetings, especially the last two evaluation conferences, were very important ways that we had a chance to disseminate our own work." (MSP Interviewee 8)

Levels of Involvement

The interviewees discussed their level of involvement in the Utah State MSP-RETA. When asked to rate their involvement, six interviewees replied that they were "involved some." Two participants suggested that they were involved "a little." In response to the question about how they were involved with the Utah State MSP-RETA, eight interviewees mentioned *professional development*. They participated in workshops in Minnesota and Washington, DC and two evaluation summits that were held in Minneapolis in September 2005 and October 2006. These participants describe their MSP's involvement in the following manner:

> …We sent teams to both Evaluation Summits that were held in Minneapolis. There was a training component to the meeting in Baltimore, training around logic models. And I attended pre-evaluation summit training led by M. Patton. That was the first year of the evaluation summit, a day-long session about mixed methods and use of qualitative methodologies. The evaluation summit number two this past October was infused with a few one-hour-long keynote presentations. (MSP Interviewee 2)

> We went to what I'm remembering to be a 2-1/2 day conference, talking about program evaluation and really doing an education for the PIs and co-PIs and program directors about sort of the breadth of evaluation… We had presentations on what is a logic model, how are logic models prepared and some general evaluation instruction and then also some individualized instruction where you signed up with one of the experts in evaluation and they actually sat with you to go through your evaluation and read through it, give advice and confer about it. (MSP Interviewee 7)

Six interviewees also indicated *evaluation consulting assistance*. It was both on-site and distance technical assistance that served as a sounding board for some of the internal evaluation work that the MSPs were already doing. As explained by MSP PIs,

> We had considerable interaction with a staff member from the Utah State RETA, and he and I had several chats about evaluation problems that we were having. He always offered constructive advice. (MSP Interviewee 5)

> We also interacted with Cathy and one consultant that was assigned to our project… The two of them came…to provide some on-site technical assistance. We also corresponded back and forth by email and got sort of distance technical assistance in addition to on-site assistance. (MSP Interviewee 2)

That assistance was related both to various *research methods* and *evaluation approaches*, helping MSPs formulate their assessment approach to the overall project, looking at a number of different strands of activities that they were engaged in. As reported by this project leader:

> They would give us feedback by saying: "Have you considered using such and such an instrument?" or…pointed me in the direction of a couple papers to look at. And we talked about what would be appropriate sample sizes for some of the many studies that we were doing. They gave us some suggestions about possibly trying stronger statistical methods like hierarchical linear modeling. (MSP Interviewee 2)

Also, two interviewees reportedly participated in the *needs assessment* facilitated by the Utah State MSP-RETA. Another way of involvement was *community building* – networking and sharing:

> So there have been a number of projects that I have come into contact with as a result of my association with the RETA conferences or workshops that allowed me to communicate and have further contact with other projects that are very similar to ours and we could share information and some research techniques and ideas. (MSP Interviewee 3)

Perceived Effect of Involvement

In terms of how the RETA used participant input, seven interviewees recognized the *consistent feedback and responsiveness* of the Utah State MSP-RETA staff. These Utah State MSP-RETA participants reported:

> As the projects moved forward, I did get calls from them asking how can we help, what are some things that would be useful to you, we can be consultants to you in various ways and I think we found them to be quite helpful… (MSP Interviewee 8)

> I feel I was asked fairly consistently whenever they were planning, so, in the planning phases of events, workshops or conferences or the summits that they have done. And also afterwards, they asked for feedback on what, as the participants, we saw was valuable and how we would suggest changes be made. So, I feel like we were consistently involved in kinds of general feedback that [they] needed or wanted. (MSP Interviewee 1)

> I think the ideas that we had were respected and when we asked for feedback on our ideas, I think they gave us pretty candid feedback. …They had some expertise that went beyond what we had but I felt like we were partners. We obviously knew more about what we were trying to do than they did. I think our relationship with the Utah State people was very collegial. (MSP Interviewee 5)

However, one person complained that in his MSP, the Utah State MSP-RETA staff *listened but were not very helpful:*

> I feel like their facilitation of the consultative relationship that we have has not been very helpful because they haven't an authority or certain level of expertise that has helped us shape our assessment program very effectively. We've struggled with that, and they haven't provided that kind of resources and expertise that we were hoping they would. They listened to us and maybe listened to us in some fashion too much because they haven't provided us with the kind of leadership that we'd needed to think of and frame our evaluation and assessment. (MSP Interviewee 4)

Perceived Use by Project Leaders and Evaluators

The interview participants rated the Utah State MSP-RETA's level of impact on them as individuals an "extensive amount" (2 persons), "some impact" (4 persons), and "a little impact" (2 persons). When asked about being personally influenced by the Utah State MSP-RETA, six interviewees reported that they *strengthened and deepened their knowledge base on evaluation.* Their appreciation of this outcome is reflected in the following quotations:

> Personally, it was an opportunity for me to learn what program evaluation is, all the way to wanting to pursue my skills in that to a much deeper extent. That had a profound impact on me personally. (MSP Interviewee 1)

> The training we had was fantastic. I really enjoyed the presentations from the different experts they brought in to the University of Minnesota and other locations. I had an opportunity to listen to some of those really world-renowned experts in the area of evaluation. (MSP Interviewee 3)

The interviewees gained some insight into how other MSP projects were evaluating

similar initiatives, and that was useful in helping decide how they would evaluate. The study participants also *learned more about STEM* and *became motivated to write and disseminate*. They generated a number of research papers as a result of the presentation papers that were made at the Utah State MSP-RETA evaluation conferences. Having the motivation to write research papers, the venue to share them and get feedback, and then be able to publish them or put them out for a broader audience was an important outcome personally for several MSP-RETA participants. Two interviewees indicated that the MSP-RETA *increased their network* of professional colleagues across the country.

Another MSP staff member noted that as a result of participation in the Utah State MSP-RETA, his project reworked the model.

> The two conferences that we presented at, which were in Minnesota, were very helpful. After the first one, we came home and reworked our model. Consequently, we were asked to present the following year what we learned and implemented as a result of our participation in the conference. (MSP Interviewee 3)

The Utah State MSP-RETA provided one person with a higher level of *confidence in evaluation expertise*. In his discussions with the Utah State MSP-RETA consultants, he gained confidence that his thoughts and ideas about evaluation of his program activities were on target. Another PI also stated that she "was able to have the confidence and the validation that I was certain that we were on a path that would be the best one that was available at the time to lead us to being an evidence-driven MSP." (MSP Interviewee 7)

Use by Projects

When asked to rate project-level impact of the Utah State MSP-RETA, three interviewees selected "extensive impact," four participants "some impact," and one person "a little impact." The interview participants described a number of ways the Utah State MSP-RETA affected their projects. Four interviewees believed that *evaluation model development* was useful, as illustrated in the following excerpt:

> Very useful in terms of any questions that we had. Again the technical assistance, the research and evaluation they provided were very useful in terms of development of our evaluation plan and then, as questions came up in the evaluation plan, some TA allowed us to modify our evaluation plan as the project developed. (MSP Interviewee 3)

> It had a large impact. We really sat down as a working group – my whole team – during the RETA session and wrote up a logic model for our project so it was used in that way of really being able to use a tool to think about common vision

> and the output at the end while we were just beginning the project… and how we were going to measure that, and the timing… (MSP Interviewee 7)

Good feedback on evaluation as an important effect of the Utah State MSP-RETA was also reported in five interviews. According to these MSP staff members,

> I think it had significant impact on how we went about doing evaluation of the project and, particularly, the support…to the project when it came to getting feedback on our evaluation plans, our data collection, models, and also feedback on some of our evaluation report drafts. (MSP Interviewee 1)

> We were given advice on evaluation questions that we had. In some cases, it helped us validate our objectives by having an expert agree with what we were trying to do. In other cases it helped one of our initiatives because the Utah State person suggested an evaluation strategy which we adopted. (MSP Interviewee 5)

Three respondents indicated *new knowledge and skills*. The MSP project management was educated about some components of evaluation that people were relatively unfamiliar with.

> The information and knowledge provided by RETA were very useful, as well as the relationships. I personally have learned many new skills. I'm not sure I can say that for everyone in our project. But I was certainly supported… Cathy, in particular, gave me some direction on ways to pursue professional development in program evaluation. (MSP Interviewee 1)

Interview participants reported that the Utah State MSP-RETA *motivated them to share evaluation results* with other partnerships and the entire field and provided valuable opportunities to present their findings and their own questions about their ongoing work. It seems that presenting at the conferences led to greater impact than just attending. As cited by two MSP project leaders,

> They helped to motivate us to assemble our earlier evaluation results and get them into a form where we could share them with our consultants through the RETA. Without them, we might have been slower in getting those things pulled together. And by virtue of pulling together these papers for the evaluation summits, that actually turned out to be useful to us to have those papers to share with other audiences that we would have been unlikely to pull them together without the RETA having prompted us to do it. (MSP Interviewee 2)

> I think it had an impact on our project by allowing us to disseminate some of our results and get feedback from various other PIs in other projects so we were able to have dialogue around our work. (MSP Interviewee 8)

One PI appreciated the fact that the external evaluator for her project was someone that she was put in touch with because of the Utah State MSP-RETA consultants:

> We had conversations about who might be good matches for evaluation that I would never have been able to do on my own. So that's the main impact is just that's how we have our external evaluator. (MSP Interviewee 6)

Factors Affecting Use

Enabling Factors. The major factors that were perceived to enable the MSP-RETA to influence survey respondents and interviewees were as follows. The *responsiveness and expertise of the Utah State MSP-RETA leadership* were mentioned by six survey respondents. The four interview respondents also praised knowledgeable, helpful, and caring people of the RETA, "a truly genuine helpfulness of the different people, just the general supportive caring about the individual MSPs and how we are doing and the questions that we were addressing throughout different stages" (MSP Interviewee 1). Another important factor was its staff availability, timeliness, and flexibility. There was timeliness to the kinds of work and opportunities that the Utah State MSP-RETA was providing. So, as MSPs were moving along in evaluation of their projects, the Utah State MSP-RETA was also providing professional development or gathering and meeting opportunities.

The survey participants appreciated "the apparent integrity and commitment of the Utah State MSP-RETA evaluator" and "regular contact with all team members and sharing of information between Utah State MSP-RETA and MSP groups." The survey open-ended responses document

> … the responsiveness of RETA leadership. Critical relationship with consultants…
> Opportunity to receive critique and recommendations…
> The extensive sharing of evaluation perspectives…

A number of survey respondents highly praised their *participation in evaluation conferences and national meetings.* The *professionalism and the expertise* that were present at the workshops had an impact on their participants. As was stated in these open-ended responses from the survey,

> I attended the evaluation conferences and found them to be the best technical assistance I could have imagined. The things that stick in my memory are: the development and use of a logic model, time to work and discuss our particular evaluation plan as a team, then time to consult with many different experts about general and specific questions.

> The thing I thought were important is that it wasn't just one evaluator that was recommending something. It was you felt like you were getting the best thinking in the field because there were many evaluators who were engaged in this dialogue around educational evaluation. So it gave it the sense of really thinking that this is the best in the field and our best thinking and the best knowledge of instruments and yet they were diverse enough they were able to address all the different questions that the project had.

According to two interviewees, the conferences were well-run and relevant to the needs of their attendees. They brought in the right kinds of people to participate and created an environment of sharing and intellectual engagement. The survey respondents who took advantage of the Utah State MSP-RETA networking opportunities also appreciated "the openness with which people dealt with their challenges to their individual projects and how different challenges influenced the questions one asked about evaluation."

Another enabling factor, *use of rigorous evaluation models and analysis,* is represented in the following survey quotations underscoring

> …the power of the analyzed data to reveal areas of strength and weakness… Use of rigorous models and statistics to help inform and guide the project. …Comprehensive - all major activities of the project evaluated both quantitatively and qualitatively.

Two more survey respondents also indicated that they valued *writing and presenting evaluation reports* at conferences as part of the Utah State MSP-RETA activities. They believed that such experiences influenced them in a positive way.

Challenging Factors. Survey respondents and interviewees suggested the following themes when asked about major factors that limited the influence the evaluation experience had on them. The *other competing priorities* theme was prominent in the responses of people who lamented about "distance over time and local workload." A number of competing priorities facing the MSP projects' management limited meeting attendance. Some of them could not attend the Utah State MSP-RETA conferences. For one interviewee, distance was a problem: "You can pick up a phone and call somebody or you can email somebody, but it's really not like face-to-face conversations" (MSP Interviewee 5).The interviewees also mentioned *limited time and resources* among the few negative factors that inhibited the impact. According to one MSP PI, there was a limit to the amount of time and intellectual investment that his project could get from the Utah State MSP-RETA (MSP Interviewee 2). Other suggestions from interviewees were to have more MSP-RETA people that were more accessible (MSP Interviewee 5), more times with the group, and follow-up (MSP Interviewee 7), as well as to provide more funds to involve MSP personnel in the Utah State MSP-RETA's activities – especially, evaluation methodology workshops (MSP Interviewee 4). It reportedly might have led to a greater impact.

A few study participants also noted *the lack of responsiveness* on behalf of the Utah State MSP-RETA consultants. It is clearly expressed in the following quotations from surveys:

> As it turns out, the evaluation consultant did very little to assist our project with the evaluation.
> …too abstract and poofy. It did not help focus what we were doing specifically on improving STEM for specific students and teachers.

According to a few MSP program leaders, *the Utah State MSP-RETA's framework was not applicable to some MSP projects.* As one survey respondent indicated, "what the Utah State MSP-RETA had to offer simply did not apply to our program." Similarly, another respondent stated that

> Our program design did not fit the typical framework that the RETA had, so conclusions regarding our fit to the framework were of limited value.

One PI believed that if the RETA had taken a *more leadership role in communicating evaluation approaches* and methods within the MSP program, that would have helped to have a greater impact:

> They seemed to have a more passive style instead of a more active and proactive style…If they had communicated more of the cutting edge approaches to evaluating math and science education activities and K-12 school performance in math and science education, even student performance related to teacher performance related to school district performance in math and science education, we would've certainly benefited from that type of leadership on their part. (MSP Interviewee 4)

The interviewees' suggestions for a greater impact also included *working from the start to the end* of MSPs' evaluation projects, as detailed in this quotation:

> I suppose if they could have worked with our project all the way to the point of having publishable results, so they would have engaged with our project earlier on at the point where we were still doing data collection and then helped us with analysis and sort of so be involved all the way through that they get us all the way to the point of having publications at the end of that. (MSP Interviewee 2)

Reportedly, it would also be useful to create some materials that MSP projects could continue to refer back to in time: "just more documentation at the end that can be referred to and that may be coming up or maybe something I'm not aware of right now but having an evaluation manual perhaps." (MSP Interviewee 7)

As additional methods to examine use of evaluation, the research team conducted two surveys to examine the broad impact of the evaluation project on the field, rather than the impact on NSF who was the intended user. The two surveys: an online survey of the STEM education field and a survey of evaluation and STEM education journal editors were designed to assess whether the MSP-RETA evaluation, its PIs, and related publications would appear among the names and titles of influential evaluation-related people, products, and activities reported by members of the STEM education and evaluation community. The researchers reasoned that if survey respondents mentioned the names and titles of the targeted program evaluation, then this might indicate the presence of a network of influence among the four targeted programs and STEM educators and evaluators.

When the research team examined the field survey results for evidence of influence of the MSP-RETA evaluation, its PIs, and related publications on the broader fields of STEM education and evaluation, they found 64 of the 191 respondents indicated they were able to identify an evaluation by name, but only 51 of the 64 respondents actually provided the name of an evaluation in a follow-up question. The MSP-RETA evaluation was named by six respondents. Four respondents named a publication associated with MSP-RETA. Finally, in response to the question regarding influential STEM evaluators, evaluators associated with MSP-RETA were not mentioned in the responses.

Focusing further, there were eight respondents who reported that they were involved in MSP-RETA, three respondents identified as influential an MSP-RETA-affiliated evaluation, publication, or evaluator by name.

As a way to further examine the presence of influence on the field, the researchers conducted an additional survey of STEM education and evaluation journals. One of the three editors who responded to the journal editor survey was able to name an influential evaluation of a large scale, multi-site STEM education evaluation but that evaluation was not the MSP-RETA program evaluation.

Project-Level Summary

This case presents evidence of the impact of the Utah State MSP-RETA program evaluation project on individual project leaders in terms of changes in knowledge, skills and beliefs of these leaders and on their future project activities.

The case findings are consistent with the notion that involvement in evaluation activities is associated with the use of the evaluation in later projects and activities. This case shows that the majority of the study respondents reportedly had some involvement in the Utah State MSP-RETA activities. The lack of evaluation capacity in their projects was a major motivating factor for them to get involved.

The interviewees were involved with the Utah State MSP-RETA in a variety of ways such as professional development (workshops and evaluation summits), community building (through networking and sharing), on-site and distance evaluation consulting assistance. The evaluation technical assistance helped MSPs enhance their evaluation approaches and research methodology for their evaluation projects. The majority of the interview participants experienced at least some level of personal impact of the Utah State MSP-RETA, which helped them to strengthen and deepen their knowledge base on evaluation, learn more about STEM, become motivated to write and disseminate, and expand their professional network.

In this case, the relationship between involvement and use was much more complex because of the different possible levels of participation; actually having an evaluation consultant work with the project versus attending a workshop. As would be expected, there appeared to be more instrumental and conceptual use when there was individual evaluator work, although the workshops appeared to result in conceptual use mostly influencing dispositions toward evaluation.

Finally, interviewees identified a number of factors that they perceived to enable the MSP-RETA to influence its participants. The survey respondents and interviewees praised responsiveness and expertise of the Utah State MSP-RETA leadership. The well-run and relevant MSP-RETA evaluation workshops and national meetings were another contributing factor, along with the use of rigorous evaluation models and analysis, and opportunities to present evaluation reports. In addition, interviewees identified factors that might have limited the influence of the MSP-RETA on its participants. These factors included other competing priorities, limited time and resources, lack of responsiveness in some cases, and the MSP-RETA's framework not applicable to some MSP projects.

Influence on the Field

This section of the case study presents data from several sources exploring the effect of the Utah State MSP-RETA on the fields of STEM education and evaluation. The different sources include the MSP-RETA project leaders' and evaluators' surveys and interviews, a citation analysis of the use of MSP-RETA materials, a survey of NSF PI's about influential evaluations and a journal editor inquiry. It is organized according to the following three topics:

- Projects' Perceptions of Influence on the Field
- The "Field's" Perceptions of Influence on the Field
- Evaluation Products' Influence on the Field

Projects' Perceptions of Influence on the Field

During interviews project leaders and evaluators were asked about the ways in which the Utah State MSP-RETA had an effect on the STEM education field. One person noted

that a major positive thing was encouraging people to do better evaluation and helping people evaluate their initiatives in such a way that the results would inform practice in STEM. For another interviewee,

> It's starting a process by which we as a community can start being much more focused on what we're evaluating, how we're evaluating it, what tools we're using and I think it's going to take a lot more collaboration but I think this is a good start. (MSP Interviewee 7)

Three respondents thought about conferences sponsored by the Utah State MSP-RETA, specifically *dissemination through conferences*. Making the research that was coming out of the MSPs available to wider audiences through the conferences and the proceedings from the conferences was valuable to these respondents. According to one of them,

> By simply having conferences where people present papers, it kind of gets projects to produce papers and consolidate their findings to analyze and present, and promotes and facilitates dissemination of very important work that's been done, in a sense. (MSP Interviewee 1)

The evaluation field, according to the interviewees, was also influenced by the Utah State MSP-RETA's work. It was made possible through *larger evaluation capacity building*. This fact is addressed in this participant's quotation:

> I think of myself as an example of someone who prior to interacting with the RETA did not know much about program evaluation. I would have learned something about it simply by working in MSP, but when we think about larger capacity building or building the knowledge of individuals who are doing evaluation, I think that that's one impact both within and then beyond the MSP community… (MSP Interviewee 1)

Another person pointed out dissemination of *new models for evaluation* in the course of the Utah State MSP-RETA project. He appreciated "a tremendous contribution and effect" caused by bringing "some fantastic experts together to present new models of conducting research and evaluation" (MSP Interviewee 3). One more interview participant was confident that the MSP-RETA contributed to agreement around what makes evaluation models successful. (MSP Interviewee 7)

Another broad impact observed by one interviewee was possibly *positive impact on NSF.* He hoped that the Utah State MSP-RETA's knowledgeable, innovative people not only had impact on individual MSPs, but also had some impact on the funding organization – the National Science Foundation. As he further explained,

> Perhaps, the Utah-RETA hopefully brought to their attention some new models that they can use to evaluate different questions and not simply rely on the traditional control-experimental approach. (MSP Interviewee 3)

For five interview participants, the most important lasting effect of the Utah State MSP-RETA on the STEM field was the *better understanding of evaluation*, which can contribute to "the quality of evaluation in general and the public's perception and understanding of what is working" (MSP Interviewee 2). Providing diverse opportunities for individuals in the field of STEM education to improve skills was important.

Because of the Utah State MSP-RETA, as one person reported, there is some concord across MSPs regarding evaluation approaches and expectations. Another respondent was pleased to find out that "there is a variety of different ways to answer questions and answer them very validly and very reliably and with some degree of assurance without having to rely on some of the old traditional models" (MSP Interviewee 3). This MSP PI observed that

> There were a lot of science and mathematics faculty members who knew almost nothing about evaluation, who had no appreciation for education evaluation. I think many of those people now have a better idea about what is involved there and I think they value it to a greater extent. I think that is a positive outcome that will be somewhat sustainable and perhaps will impact future projects. (MSP Interviewee 5)

The other important effects of the Utah State MSP-RETA mentioned in the interviews were *stronger community building*, as well as closely related *networking and exchanging ideas.* One MSP PI interviewee appreciated the opportunity to meet other MSP projects to exchange ideas with other people on different ways of thinking about how to conduct research and evaluation. He thought it would have a lasting impact on the field, the institutions of higher learning, and K-12 education. As observed by one MSP project leader in her open-ended response on the survey,

> I was initially skeptical about the involvement of RETAs in the MSP projects - mostly because, as a MSP Cohort 1, the expectations were not clearly laid out. However, following the first MSP conference, we were able to make valuable connections and it is been a wonderful and powerful experience since then.

For this person, a lasting impact was raising the awareness of the evaluation community about that content-specific nature and needs of the STEM field:

> The thing that I think they have pushed, hopefully it hasn't have gotten lost, and I think the National Science Foundation pushed this, but I think the RETA helped either to push NSF or NSF were able to use them as a vehicle to help further conversations around the country around content-specific evaluation. And that

> the evaluation needs to have content area experts involved in the work. (MSP Interviewee 6)

Another interviewee hoped that the Utah State MSP-RETA would be "a stepping stone to more proactive programs in STEM evaluation. If there is an important impact, that would be a beginning for further programs." (MSP Interviewee 4)

The "Field's" Perceptions of Influence on the Field

The research team attempted to measure the field's perception by surveying NSF PIs and journal editors. The first of the two was the NSF PI survey which gathered PIs' perceptions related to evaluations, people, people and publications. The researchers reasoned that if respondents mentioned the names and titles of the MSP-RETA program evaluation project, then this might indicate the presence of a network of influence among the four targeted programs and STEM educators and evaluators.

The research team examined the field survey results for evidence of influence of the MSP-RETA evaluation, its PIs, and related publications on the broader fields of STEM education and evaluation. Sixty-four of the 191 respondents indicated they were able to identify an evaluation by name, but only 51 of the 64 respondents actually provided the name of an evaluation in a follow-up question. The MSP-RETA evaluation was named by six respondents. Four respondents named a publication associated with MSP-RETA. Finally, in response to the question regarding influential STEM evaluators, evaluators associated with MSP-RETA were not mentioned in the responses.

Focusing further, there were eight respondents who reported that they were involved in MSP-RETA, three respondents identified as influential a MSP-RETA-affiliated evaluation, publication, or evaluator by name.

As a second look at the field's perception of influence on the field, the research team conducted a 3-question survey of STEM education and evaluation journal editors. One of the three editors who responded to the journal editor survey was able to name an influential evaluation of a large scale, multi-site STEM education evaluation and that evaluation was not the MSP-RETA program evaluation.

Evaluation Products' Influence on the Field

Our final measure of influence on the field considers the extent to which the Utah State MSP-RETA products were cited throughout the field. The citation analysis used bibliographic methods to gather data about the use of program evaluation products by individuals within the STEM education and evaluation communities. Citation analysis consists of tracking the number of citations other authors make to a published work.

This section describes the 25 products produced by the Utah State MSP-RETA evaluation technical assistance project between 2003 and 2007 and the 10 unique works which cited the projects.

The list of the RETA's products was compiled through reviewing the PI's curriculum vita, program evaluation archives, and through Internet web searches. Products were defined as any publications, presentations, instruments, or other related materials that were produced as part of the NSF-funded evaluation project. The PI herself made judgments about which of these additional products should be included, and reviewed the final list to ensure its completeness and accuracy.

The Web of Science, Google Scholar, and Google were used to collect citation information from the lists of evaluation products produced by the Utah State MSP-RETA. Searching was completed between the following dates:

Table 1. Citation search timeframe

Web of Science Searching	Google Scholar Searching	Google Searching
September 25, 2007	September 25-26, 2007	September 26, 2007

Data were collected manually by entering full reference information about the citations into an Excel spreadsheet which was later checked for accuracy and compliance with APA style and converted into an SPSS datafile.

Descriptions of the Utah State MSP-RETA Evaluation Products. The Utah State MSP-RETA evaluation technical assistance project produced a total of 25 evaluation related products. Lists of these products and the citing works to each product are presented in Appendix A. The products were coded as to their types, fields, and subfields according to the definitions presented below:

Evaluation products were coded by type according to the following:

- Instrument/tools: evaluation instruments (observation protocols, surveys, etc), instrument manuals, online evaluation tools (logic model tool, resource database);
- Presentations: conference presentations, presentations NSF or other audiences;
- Publications: journal articles, books, book chapters, published monographs, dissertations, newsletters;
- Reports: evaluation reports, briefing papers, fact sheets, brochures.

Product fields were broadly defined according to the following:

- STEM education/research: general topics or research related to STEM education;

- STEM evaluation: topics or findings connected to a specific STEM evaluation;
- Education–general: education areas/topics outside of STEM education;
- Evaluation—general: evaluation areas/issues outside of STEM evaluation;
- Other: not falling into one of the above categories.

Products were coded as to the specific sub-field within education or evaluation according to the following:

- Professional development: teacher training during service including the effects of PD on student achievement, etc.;
- Workforce development: high tech, STEM workforce training/development;
- Partnerships/systemic reform: university, school, community partnerships and systemic reform initiatives;
- Curriculum/instruction/assessment: related to classroom teaching: (i.e. instruction methods, curriculum, pedagogy, student assessment, standards-based instruction, math/science reform);
- Evaluation: evaluation-related topics such as multi-site evaluation theory or other general evaluation theory or knowledge, includes evaluation instruments/ protocols/manuals;
- Materials development: development of instructional materials;
- Project capacity building: issues related to program improvement, sustainability, advisory committee development, dissemination, replication;
- Other: other fields.

**Please note that evaluation instruments, manuals, and protocols were coded as "evaluation" rather than to what topic the instrument collected data about.*

MSP-RETA Product Types. As shown in Table 2, the majority of products produced by the Utah State RETA were presentations (n=18), with a smaller number of reports (n=4), instruments or tools (n=2), and publications (n=1).

Table 2. Types of products produced by the MSP-RETA evaluation

Product Type	N	%
Presentation	18	72.0
Report	4	16.0
Instrument/Tool	2	8.0
Publication	1	4.0
Total	*25*	*100.0*

The majority of products produced by the Utah State RETA were connected to evaluation issues facing the Math Science Partnerships or the RETA's own work (n=20), while five of the products were of a broader focus on issues in evaluation in general

(n=5). None of the RETA's products focused on educational issues specifically. Please see Table 3.

Table 3. Fields of evaluation products

Product Fields	N	%
STEM evaluation	20	80.0
Evaluation - general	5	20.0
Total	*25*	*100.0*

The Utah State RETA focused its dissemination efforts primarily on evaluation type products (n=19) and to a lesser extent on products that addressed issues in building partnerships (n=6). Details are presented in Table 4 below.

Table 4. Specific fields of evaluation products

Product Subfields	N	%
Evaluation	19	76.0
Partnerships	6	24.0
Total	*25*	*100.0*

Citations to the MSP-RETA Project Products. A total of 10 references were found to the 25 evaluation products produced by the MSP-RETA product. Ten of the eleven references were to the report: Callow-Heusser, C., Chapman, H.J., & Torres, R.T. (2005). *Evidence: An Essential Tool – Planning for and Gathering Evidence using the Design-Implementation-Outcomes (DIO) Cycle of Evidence (NSF 05-31).* The remaining citation was to the online tool: Lehman, S. (2006). *Logic Model Authoring Tool.* Logan, UT: Utah State University. [available at http://www.usu.edu/cbec]

Each of the 10 references came from different citing works; a list of the 10 citing works are presented below in Table 5.

Table 5. Works citing Utah State MSP-RETA products

Cite ID	Citation Reference Information	Type	Field	Subfield
1007	American Society for Biochemistry and Molecular Biology. (nd). K-12 Outreach Resources on the Web. Retrieved from http://www.faseb.org/asbmb/epd/orresources.html.	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment
1034	D'Avanzo, C., et al. (2006). Design and evaluation of TIEE, a peer-reviewed electronic teaching resource. Frontiers in Ecology and the Environment, 4(4), 189-195.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment

1057	Georgia Tech College of Computing - NSF Resource Office. (nd). NSF Solicitations and Important Announcements. Retrieved from http://www.cc.gatech.edu/program/NRO/.	Electronic Resource	STEM education/ research	NSF
1082	Heywood and Middleton Primary Care Trust. (2006). Draft Evidence-Based Action Plans, Healthy Lifestyles 2006-2010.	Report	Other	Other - Healthcare
1093	International Society for Computational Biology. (2005). Government reports and recommendations. Retrieved from http://www.iscb.org/weeklyDocs/june05.shtml.	Electronic Resource	Other	Other – Computational biology
1167	Nazar, B. (2006). The logic model: Past, present, and future. The center for applied management website: http://www.appliedmgt.com/eLogic%20Model.htm	Electronic Resource	Evaluation	Evaluation
1212	St. Olaf College. (nd). Survey and Questionnaire Design. Retrieved from: http://www.stolaf.edu/offices/arp/GrantEvaluations/SurveyQuestionnaireDesign.htm.	Electronic Resource	Evaluation	Evaluation
1213	Stanford University Library and Academic Information Resources. (6/20/05). Sci-Tech Library Newsletter: New e-books and reports. Retrieved from http://www-sul.stanford.edu/depts/swain/nsflibnews/2005/stan050620.html.	Electronic Resource	STEM education/ research	Other – Science and technology general resource
1223	The IDEA Partnership: National Association of State Directors of Special Education. (2007). Instructional Programs and Curricula: Science With Links to Evidence. Retrieved from http://www.ideapartnership.org/report.cfm?reportid=189.	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment
1282	Bergin, K. & Winkler, A. (2007). Challenges and Trade-offs in Measuring the Outcomes of NSF's Math and Science Partnership Program. Presentation at the MSP program regional conference, Phoenix, AZ.	Presentation	STEM evaluation	Evaluation

As is evident from analyzing the citing author list above, one of the citing works was authored by a Math Science Partnership project [ID 1282], while the remaining nine citations came from sources not affiliated with the Utah State RETA or any of the other program evaluations under consideration in the Beyond Evaluation Use project. Please see Table 6.

Table 6. Authors of citing works

Citing Work Authors	N	%
Other	9	90.0
MSP Project	1	10.0
Total	*10*	*100.0*

MSP-RETA citing works were predominantly US in origin. Nine of the citing works were from the United States, and one was from the United Kingdom.

Citing works were categorized by type according to the same categorization scheme used with the products, with the addition of codes for "electronic resources" which was used for Web pages, archived listserv messages, or similar online sources, and the category "other." As shown in Table 7, seven of the ten citing works were electronic resources, including a number of university and not-for-profit organization website resource areas. There was also one presentation, one journal article, and one report that cited the RETA's products.

Table 7. Citing work types

Types	N	%
Electronic Resource	7	70.0
Presentation	1	10.0
Publication	1	10.0
Report	1	10.0
Total	*10*	*100.0*

As shown in Table 8, half of the citing works addressed issues in STEM education. Among the five STEM education and research works, three addressed issues in curriculum, instruction, and assessment, one addressed issues in science and technology generally, and one was a request for proposals issued by NSF. The remaining five citing works addressed evaluation issues outside of STEM (n=2), and one each were related to the fields of STEM evaluation, healthcare, and computational biology.

Table 8. Citing work fields

Fields	N	%
STEM education/research	5	50.0
Evaluation	2	20.0
Other	2	20.0
STEM evaluation	1	10.0
Total	*10*	*100.0*

Citation Analysis Discussion and Conclusions. The Utah State RETA's products had limited influence on the STEM fields, with the exception of the D-I-O Model report, which has been picked up in several STEM and non-STEM areas. Interestingly, the citations to the D-I-O cycle have been primarily in collections of online resources as opposed to in written reports or articles. As the D-I-O cycle report has only been published for a few years, however, it's likely that the influence of the publication will spread and will reach journals and reports as time passes. It is somewhat surprising that the D-I-O cycle has not been cited more widely within the MSP projects as it has been a focus at a number of MSP conferences. However, this may be more a matter of it being

hard to find references to its use as opposed to it not actually being used in evaluation planning by the MSP projects.

Influence on the Field Summary

As supported by the data presented in the preceding sections, the MSP-RETA program evaluation project appears to have had limited influence on the fields of STEM education and evaluation. Though most of the survey data were not applicable to this evaluation project, some of the open-ended responses indicated evidence of influence on the field. For instance, the projects reported that the evaluation project promoted the development of a community. The MSP-RETA used its community of projects to develop an approach to evaluation, the DIO model that met the needs of NSF and allowed the community to have a common language. Many reported that they used the DIO model in current and future project activities.

The MSP-RETA project was given extra funds to enhance dissemination efforts by conducting national workshops. Despite that, when the research team examined the field survey results for evidence of influence of the MSP-RETA evaluation, its PIs, and related publications on the broader fields of STEM education and evaluation, there was only limited evidence that the MSP-RETA project had influence on the field. For instance, the MSP-RETA evaluation project was named by six respondents; four respondents named a publication associated with MSP-RETA; and PIs or evaluators associated MSP-RETA were not mentioned by any respondents.

When conducting the citation analysis to examine the possibility of influence of the MSP-RETA, The Utah State MSP-RETA's products had limited influence on the STEM fields, with the exception of the D-I-O Model report, which has been picked up in several STEM and non-STEM areas. Interestingly, the citations to the D-I-O cycle have been primarily in collections of online resources as opposed to written reports or articles. As the D-I-O cycle report has only been published for a few years, however, it's likely that the influence of the publication will spread and will reach journals and reports as time passes. It is somewhat surprising that the D-I-O cycle has not been cited more widely within the MSP projects as it has been a focus at a number of MSP conferences, however it may be more a matter of it being hard to find references to its use as opposed to it not actually being used in evaluation planning by the MSP projects.

Overall, while the MSP-RETA project personnel surveyed indicated some influence from MSP-RETA sponsored-workshops and the MSP-RETA project PI reported that the project's influence occurred mainly through involvement during various phases of evaluation planning and implementation (via expert consultation to individual projects), the NSF PI Survey and the citation analysis results show only limited influence on the fields of STEM education and evaluation.

Overall Summary: Relationship between Involvement and Use

This case study explored different aspects of the impact that the Utah State MSP-RETA had on the STEM field, MSP projects, and individuals. The case data show that the majority of the study respondents reportedly had some involvement in the Utah State MSP-RETA activities. The lack of evaluation capacity in their projects was a major motivating factor for them to get involved. The interviewees were involved with the Utah State MSP-RETA in a variety of ways such as professional development (workshops and evaluation summits), community building (through networking and sharing), on-site and distance evaluation consulting assistance. The evaluation technical assistance helped MSPs enhance their evaluation approaches and research methodology for their evaluation projects.

The majority of the interview participants experienced at least some level of personal impact of the Utah State MSP-RETA, which helped them to strengthen and deepen their knowledge base on evaluation, learn more about STEM, become motivated to write and disseminate, and expand their professional network. The project-level impact of the Utah State MSP-RETA was predominantly rated as “some impact” and “extensive impact.” Reportedly, the Utah State MSP-RETA positively affected MSP projects by means of facilitating evaluation model development, providing useful feedback on evaluation, and promoting new knowledge and skills in evaluation, and encouraging the projects to share evaluation results with other partnerships.

The Utah State MSP-RETA’s influence on the STEM field included knowledge dissemination and networking through conferences. The evaluation field was, reportedly, also influenced by the Utah State MSP-RETA’s work through evaluation capacity building. Supposedly, the most important lasting effects of the Utah State MSP-RETA on the STEM field might be improved understanding of evaluation, a stronger community of practice, networking, and the exchange of ideas among projects.

A number of factors were perceived to enable the MSP-RETA to influence its participants. The survey respondents and interviewees praised responsiveness and expertise of the Utah State MSP-RETA leadership. The well-run and relevant MSP-RETA evaluation workshops and national meetings were another contributing factor, along with the use of rigorous evaluation models and analysis, and opportunities to present evaluation reports. Among the factors that might have limited the influence of the MSP-RETA on its participants were other competing priorities, limited time and resources, lack of responsiveness in some cases, and the MSP-RETA’s framework not applicable to some MSP projects.

Appendix A: Utah State MSP-RETA Products

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
MSP001	Julnes, Patria de Lancer. (2003). Accountability assessment and the math and science partnership program. PA Times, 26(7), 12.	Publication	STEM evaluation	Partnerships/ systemic reform	0	
MSP002	Callow-Heusser, C., Chapman, H.J., & Torres, R.T. (2005). Evidence: An Essential Tool – Planning for and Gathering Evidence using the Design-Implementation-Outcomes (DIO) Cycle of Evidence (NSF 05-31). Prepared for the National Science Foundation by The Consortium for Building Evaluation Capacity, Utah State University.	Report	STEM evaluation	Evaluation	9	1007, 1034, 1057, 1082, 1093, 1212, 1213, 1223, 1282
MSP003	Callow-Heusser, C.A., Dorward, J., Lehman, S., and Chapman, H. (2007). Final Report: NSF MSP-RETA Evaluation Capacity Building Project, NSF Award Number 0233382. Washington, DC: National Science Foundation.	Report	STEM evaluation	Partnerships/ systemic reform	0	
MSP004	Worthen, B., Callow-Heusser, C.A., Dorward, J., and Lehman, S. (2004). Annual Report: NSF MSP-RETA Evaluation Capacity Building Project, NSF Award Number 0233382. Washington, DC: National Science Foundation.	Report	STEM evaluation	Partnerships/ systemic reform	0	
MSP005	Worthen, B., Callow-Heusser, C.A., Dorward, J., and Lehman, S. (2005). Annual Report: NSF MSP-RETA Evaluation Capacity Building Project, NSF Award Number 0233382. Washington, DC: National Science Foundation.	Report	STEM evaluation	Partnerships/ systemic reform	0	
MSP006	Dorward, J., Chapman, H. J., & Callow-Heusser, C. A. (2005, October). An Online Logic Model Tool and the Design-Implementation-Outcomes Cycle of Evidence: Improving Evaluation Efforts in the Math and Science Partnership Program. Paper presented at the American Evaluation Association Annual Conference, Toronto, Ontario, Canada.	Presentation	STEM evaluation	Evaluation	0	

MSP007	Chapman, H. J., Callow-Heusser, C. A., & Dorward, J. (2005, September). Establishing Partnerships to Provide Evaluation Technical Assistance and Promote Evidence-Based Designs. Paper presented at the NSF's MSP Evaluation Summit, Minneapolis, MN.	Presentation	STEM evaluation	Partnerships/ systemic reform	0	
MSP008	Callow-Heusser, C. A. (2005, January). Using evidence in Math-Science Partnerships: The DIO Cycle of Evidence. Paper presented at the National Science Foundation's MSP Learning Network Conference, Washington, DC.	Presentation	STEM evaluation	Evaluation	0	
MSP009	Callow-Heusser, C. A., Dorward, J., Webb, N., Blank, R. (2004, November). If we build it, will they come? Insights from three NSF RETA projects funded to provide evaluation assistance to MSP projects. Panel session presented at the American Evaluation Association annual conference, Atlanta, GA.	Presentation	STEM evaluation	Evaluation	0	
MSP010	Callow-Heusser, C.A., & Dorward, J. (2004, November). Providing Evaluation Technical Assistance through Nationally-Known Evaluation Experts. Presentation at the American Evaluation Association Conference.	Presentation	Evaluation - general	Evaluation	0	
MSP011	Blank, R. (2004, November). Assisting MSPs with Empirical Data to Evaluate Professional Development and Quality of Instruction. Presentation at the American Evaluation Association Conference.	Presentation	STEM evaluation	Evaluation	0	
MSP012	Webb, N. (2004, November). Adding Value to the MSP Evaluations Project. Presentation to the American Evaluation Association Conference.	Presentation	STEM evaluation	Evaluation	0	
MSP013	Callow-Heusser, C.A., & Dorward, J. (2005, November). USU's Network for Evaluation Technical Assistance: Providing Evaluation Technical Assistance to NSF's Math-Science Partnerships. Presentation at the American Evaluation Association Conference.	Presentation	STEM evaluation	Evaluation	0	
MSP014	Shapiro, N., & Frechtling, J. (2005, November). Learning Together About Doing and Evaluating K-16 Reform Efforts. Presentation at the American Evaluation Association conference.	Presentation	STEM evaluation	Evaluation	0	
MSP015	Bocian, B., & Torres, R. (2005, November). Key Features of Collaboration between Internal and External Evaluators. Presentation at the American Evaluation Association conference.	Presentation	Evaluation - general	Evaluation	0	

MSP016	Quinn, R., & Lawrenz, F. (2005, November). Making the Unwieldy Manageable: Focusing the Evaluation of a Multi-faceted Initiative. Presentation at the American Evaluation Association conference.	Presentation	Evaluation - general	Evaluation	0	
MSP017	Callow-Heusser, CA. (October 2004). Introduction to the Evidence Framework: What evidence do you look for? Presentation at the MSP Program Meeting, Evidence: an Essential Tool, Washington, DC, Oct. 21-22, 2004	Presentation	STEM evaluation	Evaluation	0	
MSP018	Callow-Heusser, CA. (October 2004). Synergistic Evidence Gathering and Use for Multiple Evaluation Purposes. Presentation at the MSP Program Meeting, Evidence: an Essential Tool, Washington, DC, Oct. 21-22, 2004	Presentation	STEM evaluation	Evaluation	0	
MSP019	Callow-Heusser, C. A., and Schneider, S. H. (2004, January). Developing logic models for MSP projects. Workshop presented at the National Science Foundation's Math and Science Partnership Learning Network Conference, Washington, DC.	Presentation	STEM evaluation	Evaluation	0	
MSP020	Worthen, B, Dorward, J, Lehman, S & Callow-Heusser, C. (January 2004). Evaluation Capacity Building for STEM/MSP Projects: A 3-Year Research, Evaluation, and Technical Assistance (RETA) Project. Slide presentation from the MSP PI Conference, Washington, D.C., January 2004.	Presentation	STEM evaluation	Evaluation	0	
MSP021	Callow-Heusser, C. A. (2003, November). Developing implementation evaluation models to provide assistance to the National Science Foundation's Math Science Partnerships. Paper presented at the American Evaluation Association's annual conference, Reno, NV.	Presentation	STEM evaluation	Evaluation	0	
MSP022	Dorward, J. (2003, November). Overview of the Challenges in Evaluating Math and Science Partnerships. Presentation to the American Evaluation Association conference.	Presentation	STEM evaluation	Evaluation	0	
MSP023	Callow-Heusser, C. A., & Engle, M. (2003, October). MSP partnerships: Defining and measuring partnerships and collaborations. Workshop presented at the National Science Foundation's workshop, Building Evaluation Capacity: Helping YOU Determine if Your MSP is "Working", Baltimore, MD.	Presentation	STEM evaluation	Partnerships/ systemic reform	0	

MSP024	Callow-Heusser, C. A. (2006). Evaluation Resources Database. Logan, UT: Utah State University. [available at http://www.usu.edu/cbec]	Instrument/ Tool	Evaluation - general	Evaluation	0	
MSP025	Lehman, S. (2006). Logic Model Authoring Tool. Logan, UT: Utah State University. [available at http://www.usu.edu/cbec]	Instrument/ Tool	Evaluation - general	Evaluation	1	1167

Endnotes

[1] Project "leaders" refers to project primary investigators, co-primary investigators, coordinators, directors, administrators, or project consultants.

[2] National Science Foundation Office of Inspector General. "Audit of NSF's Math and Science Partnership Program." OIG 04-2-003. May 14, 2004.

3 NSF. 2006 MSP Program Solicitation. http://www.nsf.gov/funding/pgm_summ.jsp?pims_id=5756

[4] NSF. 2006 MSP Program Solicitation. http://www.nsf.gov/funding/pgm_summ.jsp?pims_id=5756

[5] National Science Foundation Office of Inspector General. "Audit of NSF's Math and Science Partnership Program." OIG 04-2-003. May 14, 2004.

[6] NSF. MSP At a Glance. http://nsf.gov/pubs/2005/nsf05069/nsf05069.pdf. p.12.

[7] NSF. MSP At a Glance. http://nsf.gov/pubs/2005/nsf05069/nsf05069.pdf. p.12.

[8] National Science Foundation Office of Inspector General. "Audit of NSF's Math and Science Partnership Program." OIG 04-2-003. May 14, 2004.

[9] NSF. Teacher Institutes: A Special Report. At www.nsf.gov/news/special_reports/teacher/

[10] NSF. Math Science Partnership Program National Impact Report. http://www.nsf.gov/news/newsmedia/msp_impact/final_msp_impact_report.pdf

[11] Building Evaluation Capacity of STEM Projects. Accessed at http://be.mspnet.org/

[12] NSF. "Evidence: An Essential Tool Planning for and Gathering Evidence Using the Design-Implementation-Outcomes (DIO) Cycle of Evidence." The Consortium for Building Evaluation Capacity, April 2005.

[13] http://be.mspnet.org/index.cfm/showcase,

[14] Catherine Callow-Heusser's "Twenty Minutes"

[15] "PI Reflections"